Acoustic Fault Screening for Small Industrial Fans

This paper examines a practical screening method for detecting incipient mechanical faults in small industrial fans using airborne acoustic signals. The approach combines time-domain envelope statistics with spectral band ratios to flag anomalies without requiring contact sensors. We report simulated observations from 240 fan units across 4 sites, collected over 12 weeks. The study distinguishes screening performance from causal diagnosis, emphasizing that acoustic flags indicate candidates for inspection rather than confirmed root causes. All numeric results are presented as simulated indicators to illustrate the workflow and decision thresholds.

Supporting records are presented in Table 4.

Supporting records are presented in Figure 1.



Figure 1: Simulated program cost and output summary

1 Screening Framework and Signal Processing

The screening framework processes short acoustic recordings captured at a fixed distance from each fan housing. Signals are band-pass filtered and segmented into overlapping windows; for each window we compute crest factor, kurtosis, and the ratio of high-frequency to low-frequency energy. A two-stage classifier first applies robust z-scores against a baseline built from known-good units, then a lightweight decision tree refines borderline cases. This design prioritizes low false-negative rates for screening while keeping computational cost modest for edge deployment.

1.1 Feature Extraction and Baseline Construction

Baseline construction uses recordings from fans confirmed healthy by maintenance records. We extract features in the time and frequency domains, then normalize per unit to reduce sensitivity to mounting and load variations. The baseline is updated incrementally as new healthy data arrive, with a forgetting factor to accommodate gradual drift. Feature stability is assessed via intra-class correlation across repeated sessions; features with low stability are

down-weighted. This step is critical because small fans exhibit broad unit-to-unit variability even within the same model.

1.2 Threshold Selection and Alert Logic

Alert thresholds are derived from the distribution of baseline scores using a percentile-based rule, supplemented by a persistence requirement to suppress transient noise. An alert is raised only when the score exceeds the threshold for consecutive windows. We tuned the persistence length and percentile using simulated data to balance detection latency against nuisance alerts. The alert logic parameters and their qualitative rationale are summarized in the accompanying table. The logic intentionally favors early, low-cost inspection over immediate shutdown.

Supporting records are presented in Table 1.

Parameter	Setting	Rationale	Adjustment
Score threshold	High percentile	Limit nuisance alerts	Site-specific
Persistence	Consecutive windows	Suppress transients	Tune by noise
Baseline update	Slow forgetting	Adapt to drift	Monitor stability
Alert routing	Maintenance queue	Fit workflow	Integrate tickets

Table 1: Alert logic parameters and qualitative rationale



Figure 2: Simulated screening outcomes by fault category

2 Field Validation and Operational Integration

Field validation involved deploying the screening prototype at 4 sites and comparing alerts against maintenance findings over 12 weeks. Alerts were logged and routed to maintenance coordinators, who recorded inspection outcomes. The integration aimed to fit existing workflows without adding specialized hardware. We also tracked response times and alert-to-inspection intervals to assess operational feasibility. The following subsections detail the validation protocol and the observed outcomes, with explicit caveats about the simulated nature of the data.

Supporting records are presented in Figure 2.

Supporting records are presented in Table 2.

Aspect	Current	Desired	Gap
Alert routing	Email	Ticket system	API needed
Inspection timing	Ad hoc	Scheduled	Policy
Data storage	Local	Central	Infrastructure
Training	Brief	Formal	Materials

Table 2: Operational integration considerations

2.1 Validation Protocol and Data Collection

The validation protocol specified recording schedules, sensor placement, and inspection procedures. Each site received a handheld recorder and a simple app that guided technicians through a 12-week campaign. Inspections were triggered by alerts or by routine schedules, and findings were categorized as bearing wear, imbalance, looseness, or no fault. Data quality checks included signal-to-noise ratio thresholds and rejection of recordings with excessive background noise. The protocol steps and responsibilities are outlined in the accompanying table.

Supporting records are presented in Table 3.

Step	Action	Responsible	Timing
Baseline	Record healthy fans	Technician	Week one
Screening	Weekly recordings	Technician	Weeks two to end
Alert review	Confirm and route	Coordinator	Within 2 days
Inspection	Perform check	Maintenance	Within 5 days
Outcome log	Record findings	Technician	After inspection

Table 3: Validation protocol steps and responsibilities

2.2 Observed Screening Outcomes and Limitations

Across the simulated campaign, the screening method flagged a subset of units for inspection, with a mix of confirmed faults and nuisance alerts. Detection was more consistent for bearing-related faults than for subtle imbalance. We emphasize that these observations are simulated and do not establish causal links between acoustic features and specific failure mechanisms. Limitations include small sample size per site, varying background noise, and the absence of long-term follow-up. Outcomes by fault category are compared in the accompanying table.

Year	Output	Cost	Balance
2022	76.97	50.03	26.94
2023	95.43	62.03	33.40
2024	109.07	70.90	38.17
2025	121.61	79.05	42.56

Table 4: Simulated annual operating indicators for the screening program